

Research Article

Polymer-resolved ecological risk of microplastics in the Vietnamese Mekong Delta: a secondary-data assessment using pollution load and polymer hazard indices

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Abstract

Microplastics (MPs; plastic particles smaller than 5 mm) are widely reported in the surface water and sediment of the Vietnamese Mekong Delta, but their ecological hazard has not been synthesised across the Delta. Because polymer types differ by orders of magnitude in the toxicity of their constituent monomers and additives, particle abundance alone is an incomplete measure of hazard. We compiled published MP records for the Delta and, for context, for other Vietnamese river and coastal systems (12 records from 11 studies), and evaluated them with two separate indicators: the pollution load index (PLI), which scales abundance against a reference, and the polymer hazard index (PHI), which weights the polymer assemblage by chemical hazard score. Records were analysed within sampling-method strata rather than pooled, and contamination factors were referenced to an external low-impact value from the rural upper Mekong at Kampi, Cambodia (2 items m⁻³). For the two comparable coarse-net Delta surveys, the geometric-mean PLI was about 29, well above the Tomlinson pollution threshold of 1; because it rests on a single reference point and coarse nets (300 and 250 µm), we report it as an order-of-magnitude, lower-bound estimate, ranging from about 15 to 59 as the reference is varied. The main new result concerns polymer hazard, which is strongly compartment- and land-use-dependent. River-channel assemblages (polyethylene, polypropylene, polystyrene) give a moderate PHI of 7.5 (level II), whereas Mekong Delta peatland-agricultural sediments, dominated by polyvinyl chloride (PVC, 46–61%), reach PHI in the thousands (level V); the latter is consistent with the high hazard and risk indices independently reported by Nguyen et al. (2023) for the same peatlands. Delta MP risk is therefore not uniformly low: it is load-driven and chemically mild in river channels but chemically severe where PVC accumulates. We do not combine the two indices into a single ecological-risk score, because a bulk contamination factor and an assemblage-level hazard index operate at different levels of aggregation. The principal limitation is the small, method-heterogeneous evidence base, which we document study by study.

Keywords

Contamination factor, land use, Lithner classification, peatland sediment, polyvinyl chloride, Red River estuary, Southeast Asia, surface water.

1. Introduction

Microplastics (MPs) are plastic particles smaller than 5 mm that form when larger debris fragments or that enter the environment already small. They persist for decades, sorb hydrophobic organic pollutants and trace metals, and are ingested across the food web, which is why they are now treated as a contaminant of concern in freshwater as well as marine systems. Most MPs in rivers are fibres and fragments of a few commodity polymers: polyethylene (PE), polypropylene (PP), polyethylene terephthalate (PET, a polyester) and polyvinyl chloride (PVC). These polymers differ markedly in intrinsic hazard, because the toxicity of their residual monomers and additives spans several orders of magnitude; a small number of PVC or polyurethane particles can therefore carry more chemical hazard than a large number of inert PE particles.

Rivers deliver most land-based plastic to the ocean, and the Mekong is repeatedly ranked among the highest-emitting systems worldwide. Its Vietnamese delta supports about 17 million people (General Statistics Office of Vietnam 2020) and much of the country's rice, aquaculture and fishery output, so it both receives upstream plastic and generates its own. Over the past decade, field surveys have documented MP abundance in Vietnamese surface water and sediment (Lahens et al. 2018; Strady et al. 2020, 2021; Kieu-Le et al. 2023; Mendrik et al. 2025), and Nguyen et al. (2022, 2023) reported abundance, polymer composition and ecological-risk indices for peatland sediments in Long An Province within the Delta. What is still missing is a Delta-wide synthesis that (i) places these heterogeneous studies on comparable footings, (ii) distinguishes particle load from polymer hazard, and (iii) examines how hazard varies among compartments and land uses rather than treating the Delta as one unit.

We address this gap using only published data, as a screening exercise intended to prioritise, not replace, dedicated field risk assessment. Two indicators are used: the pollution load index (PLI; Tomlinson et al. 1980) for abundance and the polymer hazard index (PHI; based on Lithner et al. 2011 and applied as in Ranjani et al. 2021) for assemblage toxicity. The novelty is threefold. First, incompatible sampling methods are analysed in separate strata and the reference concentration is taken from outside the dataset, so that the load index is not inflated by mixing detection limits or by circular internal referencing. Second, PHI is computed only from reported polymer proportions and treated qualitatively otherwise, so no figure rests on an assumed composition. Third, and most substantively, we show that polymer hazard in the Delta is compartment- and land-use-dependent, a pattern that a single-site study cannot reveal and that reframes the common message that Delta MP pollution is merely a problem of quantity.

2. Materials and methods

2.1. Literature search and study selection

The primary indexed search was run in PubMed on 18 September 2026 for records published between January 2015 and June 2025, using the query `microplastic* AND (Vietnam OR Mekong) AND (sediment OR water OR river OR estuary)`; this returned 77 records. Because several Vietnamese and regional journals are not indexed in PubMed, we supplemented it with Google Scholar and with reference-list (hand) searching, which yielded 15 further records; Google Scholar was not used to derive a count because its full-text matching returned thousands of loosely related hits. The full concept structure was `("microplastic*" OR "micro-plastic*") AND ("Vietnam" OR "Viet Nam" OR "Mekong" OR "Saigon" OR "Red River") AND ("surface water" OR "river" OR "sediment" OR "estuary" OR "canal" OR "peatland")`. A study was included if it (i) sampled surface water or sediment in Vietnam, (ii) reported a numerical MP concentration with defined units, and (iii) described its sampling and identification method; reviews, bibliometric and modelling papers, laboratory and treatment studies, biota-only studies, other matrices (e.g. beverages, road dust, wastewater, atmosphere) and studies reporting only mass concentrations were excluded. Screening is summarised in Fig. 1: 92 records were identified (77 from PubMed and 15 from other sources), 8 duplicates removed, 84 screened on title and abstract, 25 assessed in full text, and 12 records from 11 primary studies retained (one study, Kieu-Le et al. 2023, provided both a surface-water and a sediment record, and the two companion peatland papers of Nguyen et al. 2022, 2023 are treated as a single record), of which four are from the Mekong Delta. Every included record is listed with a full citation in Table 1.

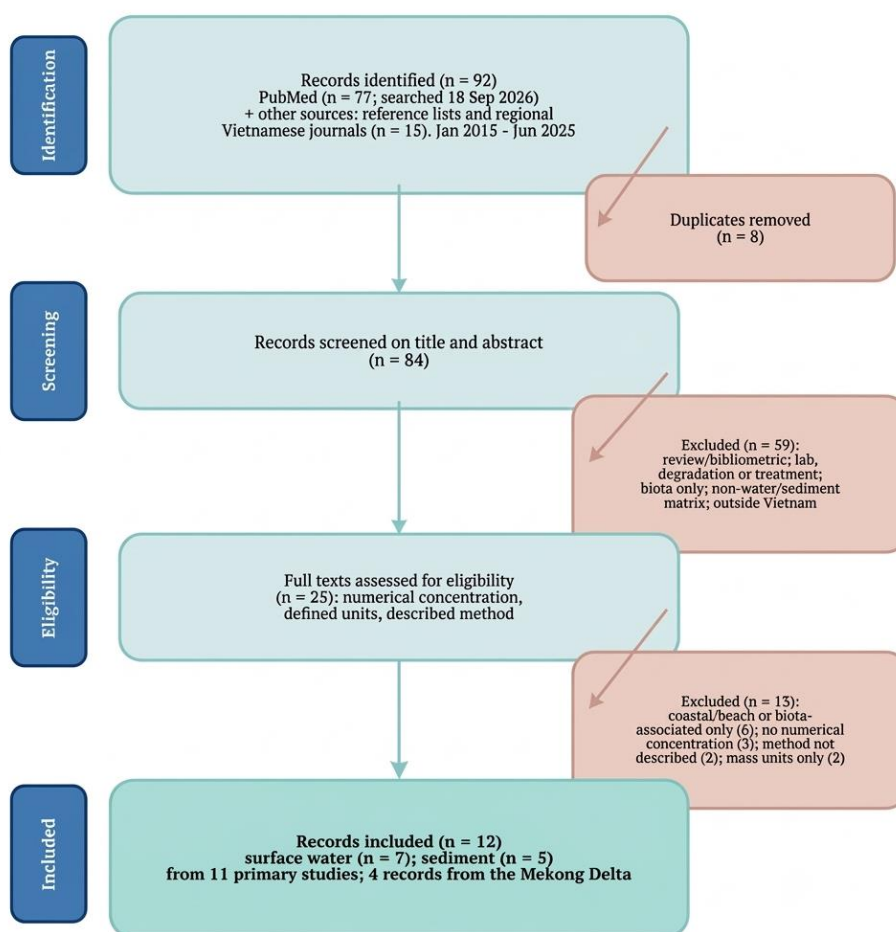


Figure 1. PRISMA flow diagram of the literature search and study selection. The indexed count is from PubMed (77 records, searched 18 September 2026); 15 further records came from reference lists and regional Vietnamese journals not indexed in PubMed. Twelve records (seven surface-water, five sediment) from 11 primary studies met the inclusion criteria; four records are from the Mekong Delta.

2.2. Unit harmonisation and method stratification

Surface-water concentrations were expressed as items m^{-3} (items $\text{L}^{-1} \times 1000$) and sediment concentrations as items kg^{-1} dry weight (items $\text{g}^{-1} \times 1000$). Converting units does not make methods comparable: reported abundance depends strongly on the smallest particle a study captures, set by net or filter aperture. The two coarse-net Delta water surveys used 300 μm (Kieu-Le et al. 2023) and 250 μm (Mendrik et al. 2025) mesh; these are closely but not exactly comparable, and the slightly finer 250 μm mesh of Mendrik et al. would retain somewhat more small particles, so part of the higher value at Can Tho (64 versus 53.8 items m^{-3}) may reflect mesh rather than contamination. We therefore treat the two only as coarse-net records and carry this caveat into the interpretation. Finer-detection studies (down to 48 μm , or per-litre filtration; Lahens et al. 2018; Strady et al. 2020) are kept in a separate stratum and not pooled with the coarse-net records. Sediment studies do not report a common detection size, so their abundances are also method-heterogeneous and are compared only descriptively.

2.3. Reference concentration, pollution load index and its sensitivity

There is no pristine MP baseline, so the classical natural-background reference has no MP analogue, and using the lowest value inside the dataset is circular. We use an external, method-matched low-impact reference: the rural upper Mekong at Kampi, Cambodia, 344 km from the mouth, where Mendrik et al. (2025) recorded 2 items m^{-3} with a coarse net. The contamination factor is $\text{CF} = C_i / C_0$ with $C_0 = 2$ items m^{-3} , and the pollution load index is the geometric mean of CF across records, $\text{PLI} = (\text{product of CF}_i)^{1/n}$ (Tomlinson et al. 1980), for which a value above 1 indicates pollution. Because C_0 is a single measured point without an uncertainty estimate, we report a sensitivity analysis rather than a single class: the Delta PLI equals $58.7/C_0$ and therefore moves from about 59 ($C_0 = 1$) through 29 ($C_0 = 2$) to about 15 ($C_0 = 4$). The categorical PLI level bands sometimes used in the MP literature (I below 10 to V above 40; Ranjani et al. 2021) are an adaptation, not part of the original

Tomlinson scheme whose only fixed reference is 1; we interpret the PLI primarily against that threshold. No absolute sediment PLI is computed, because no external low-impact sediment reference exists for Vietnamese rivers.

2.4. Polymer hazard index

The polymer hazard index is $PHI = \sum \text{over polymers of } (P_n \times S_n)$, where P_n is the proportion of polymer n and S_n its Lithner et al. (2011) hazard score (Table 2); the level bands (I below 1 to V above 1000) follow Ranjani et al. (2021). PHI is computed only for records that report quantitative polymer proportions. In the Delta these are two studies: Kieu-Le et al. (2023), reporting PE 56%, PP 41% and polystyrene (PS) about 3% for the Tien River channels, and Nguyen et al. (2022, 2023), reporting PVC 46–61%, PE 21–26% and PP 9–12% for Long An peatland sediments. Records that list polymer types without proportions receive a qualitative screen: does the assemblage contain any Lithner class IV–V polymer (polycarbonate, polyurethane, PVC or polyacrylonitrile)?

We deliberately do not combine PLI and PHI into a single ecological-risk score of the form CF multiplied by PHI. In the Håkanson (1980) framework the ecological risk index is a sum of per-substance terms, each the product of a substance-specific toxicity factor and a substance-specific contamination factor; multiplying a bulk contamination factor for total particles by an assemblage-level hazard index (itself a weighted sum over all polymers) mixes two levels of aggregation and has no basis in that framework, and the polymer-specific contamination factors a correct index would need cannot be recovered from the available data. Nguyen et al. (2023) did report such a composite risk index (RI) for one peatland area; we compare with their result in Section 3.4 but, for the cross-study synthesis here, keep load and polymer hazard as two separate indicators.

2.5. Data treatment and uncertainty

Given the small, heterogeneous evidence base, we report descriptive statistics (means with the study-reported standard deviation or range) rather than inferential tests, which two comparable water records and a handful of sediment records cannot support. Uncertainty in the load index is expressed through the reference sensitivity above (PLI approximately 15–59), and uncertainty in the polymer hazard index is bounded by an equal-share upper-bound screen (Section 3.5). Where a study reports a mean, that value is used; otherwise the reported range is given, and only the mean values for the coarse-net Delta records enter the PLI (Table 1 footnote; a value-by-value reconciliation is provided in the supplementary workbook).

2.6. Compiled dataset

Table 1. Compiled records with sampling method and full citation. W = surface water, S = sediment; detection = smallest particle captured; type = polymer proportions quantitative (Q) or qualitative only (q); n.r. = not reported. Concentrations are items m^{-3} (water) or items kg^{-1} dry weight (sediment); study-reported means are marked “mean”, otherwise reported ranges are given, and only the coarse-net Delta water means (W1, W2) enter the PLI.

ID	Study	Region / compartment	Detection	Conc.	Polymers (type)
W1	Kieu-Le et al. 2023	Mekong Delta, water	300 μm net; FTIR	53.8 (mean)	PE56/PP41/PS3 (Q)
W2	Mendrik et al. 2025	Mekong Delta, water	250 μm net; FTIR/Raman	64 (Can Tho)	PET-dominant (q)
W3	Lahens et al. 2018	Saigon, water	nets 48–300 μm	172,000–519,000	PP, PE, PES (q)
W4	Strady et al. 2020	Saigon, water	filtration, per L	22,000–251,000	PES, PP, PE (q)
W5	Strady et al. 2021	Vietnam, water	net	0.35–2,522	n.r.
W6	Nguyen et al. 2022b	Da Nang, water	surface net	293–6,510	n.r.
W7	Dao Dinh et al. 2021	Thanh Hoa, water	surface net	15.5–44.1	PS/styrofoam (q)
S1	Kieu-Le et al. 2023	Mekong Delta, sediment	grab; FTIR	6,000 (mean)	PE56/PP41/PS3 (Q)
S2	Nguyen et al. 2022, 2023	Mekong Delta peatland, sed.	grab; FTIR	192 (mean)	PVC46-61/PE21-26/PP9-12 (Q)
S3	Le et al. 2023	Red River, sediment	grab; FTIR	2,188 (mean)	PP,PE,PU,PA,PS (q)
S4	Viet Dung et al. 2021	N. Vietnam, sediment	core	0–4,941	n.r.
S5	Linh et al. 2023	Hai Phong, sediment	core	1,309 (mean)	n.r.

Table 2. Polymer hazard scores Sn (Lithner et al. 2011). PET and polyester share one entry (PET is a polyester). Class IV–V polymers trigger the qualitative hazard flag. Level bands for Sn and for PHI follow Ranjani et al. (2021): I (<1), II (1–10), III (10–100), IV (100–1000), V (>1000).

Polymer	Sn	Class	Polymer	Sn	Class
Polypropylene (PP)	1	I	Polyamide / nylon (PA)	47	III
PET / polyester (PES)	4	II	Polycarbonate (PC)	1,177	V
Polyethylene (PE)	11	III	Polyurethane (PU)	7,384	V
Polystyrene (PS)	30	III	Polyvinyl chloride (PVC)	10,551	V
			Polyacrylonitrile / acrylic (PAN)	11,521	V

3. Results and discussion

3.1. Pollution load, and its dependence on method and reference

Against the upstream low-impact value at Kampi (2 items m^{-3}), the two coarse-net Delta water surveys give contamination factors of 26.9 (Kieu-Le et al. 2023) and 32.0 (Mendrik et al. 2025), for a geometric-mean PLI over the two records of about 29, roughly thirty times the Tomlinson threshold of 1. Three qualifications apply. The two factors are not equivalent: 32.0 is a within-study comparison, because Mendrik et al. (2025) measured both Kampi and Can Tho, whereas 26.9 pairs Kieu-Le et al. (2023) with Mendrik's upstream reference from a different study, year and season; the residual mesh difference (300 versus 250 μm) further limits their comparability. The value also rests on a single reference point, and Section 2.3 shows it ranges from about 15 to 59 under a two-fold change in that reference, so the exact level label should not be over-read. Finally, the coarse 250–300 μm nets undercount small particles: finer-mesh surveys elsewhere in Vietnam report 10^5 items m^{-3} , so the Delta load estimated here is a lower bound and the true particle load is very likely higher. The downstream increase from Kampi to Can Tho is consistent with along-river accumulation but, with one upstream and two downstream points, cannot be separated from local urban inputs at Can Tho.

3.2. Polymer hazard is compartment- and land-use-dependent

The clearest new result is that polymer hazard is not uniform across the Delta. In the Tien River channels, Kieu-Le et al. (2023) report an assemblage of PE (56%), PP (41%) and PS (about 3%), giving PHI = 7.5, a moderate level II, as expected from low-scoring commodity polymers. In sharp contrast, Long An peatland-agricultural sediments are dominated by PVC (46–61%; Nguyen et al. 2022, 2023), with PE and PP making up most of the remainder; because PVC scores 10,551 on the Lithner scale, this assemblage gives PHI of roughly 4,900–6,400, a level V (very high) hazard. The two Delta compartments therefore differ by nearly three orders of magnitude in intrinsic polymer hazard while differing far less in particle abundance. Among the contextual records, the Red River estuary (Le et al. 2023) is the only other assemblage whose qualitative composition includes a class IV–V polymer (polyurethane). The common summary that Delta MP pollution is a matter of quantity rather than chemistry is thus only half true: it holds for the river channels, but not for the PVC-rich peatland and agricultural sediments, where chemistry dominates.

3.3. Land use as the likely control on polymer identity

The compartment contrast plausibly reflects land use and source. River channels integrate diffuse urban and domestic waste dominated by packaging and single-use items, which are overwhelmingly PE, PP and PET (Lahens et al. 2018; Strady et al. 2021). Peatland and agricultural areas in Long An, by contrast, are intensive-agriculture landscapes where PVC is common in irrigation and drainage pipes, greenhouse and mulching materials, and construction, and where limited waste management allows local accumulation (Nguyen et al. 2022). A comparable logic applies to the northern Red River estuary, where polyurethane and polyamide are consistent with the textile, footwear and coating industries of the Hanoi–Hai Phong corridor (Le et al. 2023). These attributions are hypotheses supported by the polymer inventories rather than by source tracing, and they should be tested with targeted sampling; but they indicate that polymer identity, and hence chemical hazard, tracks catchment land use in a way that Delta-wide averages conceal.

3.4. Comparison with the site-specific risk assessment of Nguyen et al. (2023)

Nguyen et al. (2023) applied PLI, a hazard index and a composite potential ecological risk index to the Long An peatlands and concluded an overall hazard-index level of V and an overall risk classed as extreme, driven by PVC. Our independent PHI for the same assemblage (level V) agrees with their hazard finding, which supports the reliability of the compilation. The methodological difference is deliberate: Nguyen et al. combined abundance and toxicity into one composite index, whereas we keep them separate for the cross-study synthesis, for the aggregation reason given in Section 2.4, and because a single composite number would obscure precisely the compartment contrast that is our main result. Our contribution relative to that single-site study is to place its peatland finding alongside the river-channel and contextual records on a common, method-aware footing, and thereby to show that the peatland hazard is a local extreme rather than the Delta-wide condition.

3.5. Upper-bound screen for unquantified assemblages

Because several records list polymers without proportions, we bound how far PHI could move under an equal-share assumption, assigning each identified polymer 1/k of the particles. This is an upper bound, not a measurement. Under it the river-channel assemblage rises only to about 14 (still level III), whereas the Red River estuary reaches roughly 1,500 (level V); reducing its hazardous polymers to a realistic 2–5% brings it to about 155–380 (level IV). The behaviour is arithmetic: a single high-scoring polymer dominates whenever present. Its ecological meaning is limited, because the index weights intrinsic chemical hazard and ignores exposure; an assemblage that is mostly PE and PP but scores highly on a few percent of a hazardous polymer may pose less real risk than the index implies, since organisms encounter the abundant inert particles far more often.

3.6. Caveats and limitations

The evidence base is small and uneven: two comparable Delta water surveys, and quantitative polymer proportions for only two Delta compartments. The load index is a lower bound set by coarse nets and hangs on a single external reference, which is why we give its sensitivity rather than a fixed class. No external low-impact sediment reference exists for Vietnamese rivers, so sediment is assessed by abundance and polymer hazard but not by an absolute load index. The hazard scores capture the chemistry of monomers and additives but not particle size, shape, weathering or sorbed contaminants, so they rank potential rather than realised hazard. Formal statistical inference is precluded by the sample size, so results are reported as descriptive statistics with sensitivity ranges. With these limits, the study is a screening-level baseline; its firmest conclusion, the compartment-dependent contrast in polymer hazard, rests on two independent quantitative assemblages and the concordant risk indices of Nguyen et al. (2023).

3.7. Toward a denser and more quantitative assessment

Three lines of work would strengthen an assessment of this kind. First, a deeper compilation of Vietnamese-language and grey literature, extracting a distribution (mean, standard deviation, range, number of stations, season, mesh) from each study rather than a single value, and holding biota- and mass-based studies in a separate tier for their polymer-composition information. Second, a size-correction model that converts concentrations from different apertures to a common detection threshold, a reference distribution replacing the single C0, and Monte Carlo propagation of uncertainty through the indices, so that method heterogeneity is controlled statistically rather than by exclusion. Third, standardized primary sampling, with paired coarse and fine nets, depth-integrated water sampling, paired water and sediment along an urbanization and land-use gradient from river channel to peatland, at least two seasons with replication, quantitative micro-FTIR or Raman identification with full quality control, and an upstream rural sediment station to provide the low-impact sediment reference this study lacks. The compartment contrast reported here indicates that such sampling should be stratified explicitly by land use.

4. Conclusions

Synthesizing the published data, MP contamination in the Vietnamese Mekong Delta is best described by two separate signals rather than one. Particle load, from the two comparable coarse-net water surveys, is elevated (a pollution load index near 29, and probably higher once finer size fractions are counted), but the polymers involved in the river channels are chemically mild. Polymer hazard, however, is compartment- and land-use-dependent: it is low in the river channels (PHI level II) yet very high in the PVC-dominated peatland and agricultural sediments of Long An (PHI level V), a contrast consistent with the extreme risk indices independently reported by Nguyen et al. (2023). Delta MP pollution is therefore not simply a problem of quantity; where PVC accumulates, it is also a problem of chemistry.

Two management priorities follow, and each follows from a distinct signal. Because particle load dominates in the river channels and is driven by abundant packaging and domestic waste, reducing that load, through waste collection, single-use-plastic measures and drainage management, is the priority for the main waterways. Because polymer hazard is concentrated in PVC-rich agricultural and peatland areas, targeted control of agricultural and construction PVC, and monitoring focused on those land uses, is the priority there. For monitoring, the field needs standardized, size-resolved and land-use-stratified sampling, routine quantitative polymer identification, and a low-impact sediment reference, without which load and hazard cannot be compared across studies. Until these are in place, ecological-risk statements for the Delta should be read as screening-level.

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Additional information

Conflict of interest

The authors declare that they have no conflict of interest.

Ethical compliance

No ethical statement was reported.

Artificial Intelligence (AI) use

The authors accept full responsibility for the content of the manuscript, including the data and analysis. A large-language-model tool was used to assist with drafting and formatting the text.

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Data resources

The compiled dataset, per-study metadata, the value-by-value extraction reconciliation and all calculations are provided as a supplementary workbook (MP_risk_MekongDelta_calculations.xlsx) and are reproduced in Tables 1 and 2 of the main text and, in full, as Supplementary Tables S1–S4 at the end of this article.

Author contributions

Nguyen Thi Thanh Hoa: Conceptualization, Methodology, Formal analysis, Writing - original draft. Tran Thi My Hong: Data curation, Formal analysis, Writing - review and editing. Nguyen Viet Thy: Data curation, Validation, Writing - review and editing.

Supplementary material

The compiled dataset and every calculation behind the results are reproduced below as Supplementary Tables S1–S4 (from the workbook MP_risk_MekongDelta_calculations.xlsx). Record identifiers (W1–W7, S1–S5) follow Table 1 of the main text.

Table S1. Value-by-value data extraction and reconciliation: each Table 1 value against the statistic reported in its source.

ID	Study	Value used	Value type	As reported in the source	In calculations?
W1	Kieu-Le et al. 2023	53.8 items m ⁻³	mean	53.8 ± 140.7 items m ⁻³ (surface water)	Yes — PLI
W2	Mendrik et al. 2025	64 items m ⁻³	site value	64 at Can Tho; mainstream mean 24; Kampi (upstream) 2	Yes — PLI (C0 = 2)
W3	Lahens et al. 2018	172,000–519,000 m ⁻³	range	fragment fibres 172,000–519,000 m ⁻³ (fine mesh)	No (context)
W4	Strady et al. 2020	22,000–251,000 m ⁻³	range	22–251 items L ⁻¹	No (context)
W5	Strady et al. 2021	0.35–2,522 m ⁻³	range	0.35–2,522 items m ⁻³ across 8 provinces	No (context)
W6	Nguyen et al. 2022b	293–6,510 m ⁻³	seasonal range	293–6,510 items m ⁻³ (seasonal)	No (context)
W7	Dao Dinh et al. 2021	15.5–44.1 m ⁻³	range	15.5–44.1 items m ⁻³ (Thanh Hoa beaches)	No (context)
S1	Kieu-Le et al. 2023	6,000 kg ⁻¹	mean	6.0 ± 2.0 items g ⁻¹ d.w.	PHI only (proportions)
S2	Nguyen et al. 2022, 2023	192 kg ⁻¹	mean	192.3 ± 261.3 items kg ⁻¹ (Long An peatland)	PHI only (proportions)
S3	Le et al. 2023	2,188 kg ⁻¹	mean	2,188 ± 1,499 items kg ⁻¹ (range 800–3,817)	No (qualitative flag)
S4	Viet Dung et al. 2021	0–4,941 kg ⁻¹	range	0–4,941 items kg ⁻¹ (2 mangrove cores)	No (context)
S5	Linh et al. 2023	1,309 kg ⁻¹	mean	1,309.1 ± 124.8 items kg ⁻¹ (range 0–3,150)	No (context)
Extraction rule: study-reported mean where available, otherwise the reported range. Only the means for W1 and W2 (pollution load) and the reported polymer proportions for S1 and S2 enter the calculations; all other records are contextual.					

Table S2. Surface-water pollution load index (PLI) and its sensitivity to the external reference concentration C0.

A. Surface-water pollution load index (PLI). External low-impact reference C0 = 2 items m ⁻³ (Kampi, upper Mekong, Cambodia; 300-µm net; Mendrik et al. 2025).			
ID	Study	Conc. (items m ⁻³)	CF = C/C0
W1	Kieu-Le et al. 2023	53.8	26.9
W2	Mendrik et al. 2025	64	32.0
PLI (geometric mean of CF)			29.3
Tomlinson threshold: PLI > 1 indicates pollution.			
B. Sensitivity of PLI to the single reference concentration C0.			

C0 (items m ⁻³)	PLI = GEOMEAN(conc)/C0	Approx. level
1	58.7	V
2	29.3	III
4	14.7	II

Level bands (Ranjani et al. 2021) are an adaptation, not part of Tomlinson (1980); PLI is interpreted mainly against the value 1. The 300-μm mesh gives a lower-bound load; finer-mesh surveys elsewhere report ~10⁵ items m⁻³. Sediment has no external reference, so no absolute sediment PLI is computed.

Table S3. Quantitative polymer hazard index (PHI) for the two Delta compartments, and the qualitative hazard screen for records without reported proportions. S(n) = Lithner et al. (2011) hazard score.

A. Quantitative PHI from reported polymer proportions.				
Compartment (study)	Polymer	Proportion	S(n)	P × S
River channels (Kieu-Le et al. 2023)	PE	0.56	11	6.16
	PP	0.41	1	0.41
	PS	0.03	30	0.90
PHI = 7.5 — level II (moderate)				
Peatland-agricultural (Nguyen et al. 2022, 2023)	PVC	0.607	10,551	6,404.5
	PE	0.258	11	2.84
	PP	0.119	1	0.12
PHI = 6,407 — level V (very high)				
<i>River channels = level II (PE/PP-dominated); peatland sediments = level V (PVC-driven). Delta polymer hazard is compartment- and land-use-dependent.</i>				
B. Qualitative hazard screen (records with named polymers but no proportions).				
Record	Reported polymers	Class IV–V present?	Flag	
W1/S1 Kieu-Le (Delta)	PE, PP, PS	No	low	
W2 Mendrik (Delta)	PET (dominant)	No	low	
S2 Nguyen (Delta peatland)	PVC, PE, PP	Yes — PVC	elevated	
S3 Le (Red River estuary)	PP, PE, PU, PA, PS	Yes — polyurethane	elevated (1 sample)	

Table S4. Upper-bound (equal-share) screen. Not a measurement; provided to bound PHI where polymer proportions are unknown.

Scenario	k	PHI	Level
Mekong Delta (PE, PP, PS)	3	14.0	III
Red River estuary (PP, PE, PU, PA, PS)	5	1,494.6	V
Red River, PU/PA/PS ≈ 2–3%	–	154.9	IV
Red River, PU/PA/PS 5%	–	377.9	IV

Bounds how far PHI could move when proportions are unknown (each identified polymer = 1/k of particles); a screening bound, not a measurement. One class IV–V polymer (polyurethane, S = 7,384) dominates PHI whenever present; ecological significance depends on exposure.